

SAM A. BNMORAAD

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New York City, NY

Computer Scientist

I am a results-driven software engineer with a passion for backend systems, data optimization, and scalable solutions. My background spans financial institutions, postal logistics, and academic research, with a consistent record of improving system efficiency, ensuring compliance, and enabling better decision-making through data.

Areas of Expertise

Backend Development • API Optimization • Microservices Architecture • Data Analysis • PostgreSQL Optimization • OAuth2/JWT • Docker & Kubernetes • Scientific Computing • Medical Software Tools • Research & Technical Writing • Multilingual Communication (English, French, Arabic)

Education

Bachelor of Science in Computer Science | Columbia University – New York City, NY

May 2027

Professional Work History

2025 Software and Website Development Intern | Columbia College IT

January 2024 – May 2024

- Migrated 10–15 Columbia departmental websites from Drupal 7 to Drupal 10, writing migration scripts, refactoring custom modules, and handling regression testing, improving average page load times by ~35% and eliminating 100% of deprecated modules in production
- Integrated a Django + PostgreSQL + RESTful API backend with a new Columbia admissions website, scaling reliably to 50,000+ applicants per cycle, cutting form submission errors by ~40%, and reducing manual data correction work by ~50%

Software Engineer | Tunisian Postal Office – Remote

April 2023 – September 2024

- Led optimization of parcel tracking APIs and contributed to a microservices-based system redesign for higher scalability and modularity
- Integrated OAuth2 and JWT for secure access management, reducing unauthorized access incidents
- Reduced API latency by 40% and improved tracking accuracy by 25%

Freelance Software Developer | Fiverr – Remote

November 2023 – May 2024

- Delivered customized software and cybersecurity solutions for 50+ clients, including API integration, mobile/web app development, and performance tuning

Finite Element Analysis Research | Lehigh Collaboration

November 2021 – December 2021

- Designed C++ plugins for ABAQUS to simulate cranial bone remodeling post-trauma.
- [Recognized as a Peer Reviewer for Annals of Internal Medicine](#) for outstanding contribution.

Freelance Software Developer | Fiverr – Remote

November 2023 – May 2024

- Developed an automated Python benchmarking toolkit for evaluating large language model (LLM) responses in LGBTQ+ suicide prevention, contributing to a peer-reviewed publication on ethical alignment and emotional intelligence in AI-driven mental health chatbots.

Technical Skills

Data Programming Languages: Python, C++, Java, JavaScript, SQL, HTML/CSS, Ruby

Backend & DevOps: Django, RESTful APIs, OAuth2, JWT, Docker, Kubernetes

Data Analysis & Databases: PostgreSQL, ABAQUS, ANSYS, MATLAB, R

Scientific & Analytical Tools: Finite Element Modeling, Simulation Software

Web & App Development: API Integration, Web Optimization, Microservices Architecture

Languages: English (Native), French (Fluent), Arabic (Fluent)